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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 38.3

$$\begin{aligned}
 I &= \int \frac{2x+1}{4x^2-4x+3} dx \\
 &= \int \frac{\frac{1}{4}(8x-4)+2}{4x^2-4x+3} dx \\
 &= \frac{1}{4} \int \frac{8x-4}{4x^2-4x+3} dx + 2 \int \frac{1}{4x^2-4x+3} dx
 \end{aligned}$$

$$\text{Stel: } t = 4x^2 - 4x + 3 \Rightarrow dt = (8x - 4) dx$$

$$\begin{aligned}
 &= \frac{1}{4} \int \frac{dt}{t} + 2 \int \frac{1}{4x^2-4x+3} dx \\
 &= \frac{1}{4} \ln |t| + 2 \int \frac{1}{(2x-1)^2+2} dx \\
 &= \frac{1}{4} \ln |4x^2-4x+3| + 2 \int \frac{1}{(2x-1)^2+2} dx
 \end{aligned}$$

$$\text{Stel: } u = 2x - 1 \Rightarrow du = 2dx \Rightarrow dx = \frac{du}{2}$$

$$\begin{aligned}
 &= \frac{1}{4} \ln |4x^2-4x+3| + 2 \int \frac{1}{u^2+2} \cdot \frac{du}{2} \\
 &= \frac{1}{4} \ln |4x^2-4x+3| + \int \frac{1}{u^2+(\sqrt{2})^2} du \\
 &= \frac{1}{4} \ln |4x^2-4x+3| + \frac{1}{\sqrt{2}} \text{Bgtan} \left(\frac{u}{\sqrt{2}} \right) + c \\
 &= \frac{1}{4} \ln |4x^2-4x+3| + \frac{1}{\sqrt{2}} \text{Bgtan} \left(\frac{2x-1}{\sqrt{2}} \right) + c
 \end{aligned}$$

References